

Serial Binary Adder

1. Introduction

A **Serial Binary Adder** is a sequential digital circuit that performs binary addition one bit at a time, beginning from the Least Significant Bit (LSB). Unlike parallel adders, which use multiple full adders simultaneously, a serial binary adder uses only one full adder and processes each bit over multiple clock cycles.

This method is slower but significantly reduces hardware complexity, making it ideal for systems where minimizing cost, power, and area is crucial.

2. Key Components

- **Shift Registers:** Two shift registers store the binary numbers A and B . At each clock cycle, the bits are shifted to bring the next pair of bits into the full adder.
- **Full Adder:** A single full adder performs the bit-wise addition of the LSBs from the two registers, along with the carry-in from the previous cycle.
- **Carry Flip-Flop:** Stores the carry-out from the full adder and feeds it as carry-in for the next clock cycle.
- **Result Register:** Collects the sum bits produced in each cycle to form the final result.

3. Working Principle

Let two 4-bit binary numbers be:

$$A = A_3A_2A_1A_0, \quad B = B_3B_2B_1B_0$$

The step-by-step operation of a serial adder is:

1. Initialization:

- Load A and B into their respective shift registers.
- Set initial carry $C_{in} = 0$.

2. Serial Addition (for n clock cycles):

- Add the LSBs of A and B along with C_{in} using the full adder.
- Store the sum bit into the result shift register.
- Save the carry-out (C_{out}) into the carry flip-flop for the next cycle.
- Right shift both shift registers by one bit.

3. Completion:

After n clock cycles, the addition is complete. If the final carry-out is 1, it is appended as the MSB of the result.

4. Full Adder Logic

At each clock cycle, the full adder performs:

$$\text{Sum (S)} = A \oplus B \oplus C_{in}$$

$$\text{Carry Out (Cout)} = (A \cdot B) + (B \cdot C_{in}) + (A \cdot C_{in})$$

This logic ensures correct bit-wise addition with carry propagation.

5. Example

Add $A = 1011$ and $B = 1101$ using a serial adder:

Cycle	A Bit	B Bit	Cin	Sum	Cout
1	1	1	0	0	1
2	1	0	1	0	1
3	0	1	1	0	1
4	1	1	1	1	1
5	-	-	1	1	0

Result: 110000 (6 bits including final carry)

6. Serial Binary Adder – Block Diagram

The following components are interconnected to build a serial adder:

- Shift Register A → Full Adder Input A
- Shift Register B → Full Adder Input B
- Carry Flip-Flop → Full Adder Carry In
- Full Adder Sum → Result Shift Register
- Full Adder Carry Out → Carry Flip-Flop

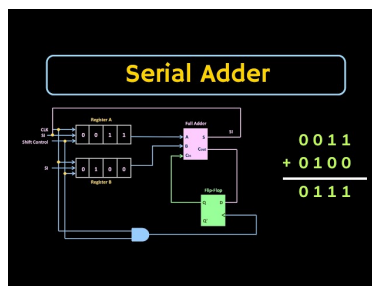


Figure 1: Block Diagram of Serial Binary Adder

7. Advantages

- Requires minimal hardware — only one full adder is needed.
- Lower power consumption and cost.
- Compact and suitable for embedded or resource-constrained systems.

8. Disadvantages

- Slower than parallel adders due to sequential processing.
- Requires n clock cycles for an n -bit addition.

9. Applications

- **Embedded Systems:** Ideal for low-power and area-constrained applications.
- **Serial Communication:** Used where data is transmitted and processed bit by bit.
- **Simple ALUs:** Incorporated in microcontrollers and simple processors where speed is not critical.

10. Conclusion

The Serial Binary Adder is a core example of sequential arithmetic logic, demonstrating how complex operations can be performed efficiently over time using minimal hardware. Despite its slower operation, it remains crucial in domains that prioritize simplicity, power efficiency, and reduced silicon area over speed.